



A spectral-ensemble deep random vector functional link network for passive brain-computer interface

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ABSTRACT

Randomized neural networks (RNNs) have shown outstanding performance in many different fields. The superiority of having fewer training parameters and closed-form solutions makes them popular in small datasets analysis. However, automatically decoding raw electroencephalogram (EEG) data using RNNs is still challenging in EEG-based passive brain-computer interface (pBCI) classification tasks. Models with the high-dimension input of EEG may suffer from overfitting and the intrinsic characteristics of non-stationary, high-level noises and subject variability could limit the generation of distinctive features in the hidden layers. To address these problems in EEG-based pBCI tasks, this work proposes a spectral-ensemble deep random vector functional link (SedRVFL) network that focuses on feature learning in the frequency domain. Specifically, an unsupervised feature-refining (FR) block is proposed to improve the low feature learning capability in RNNs. Moreover, a dynamic direct link (DDL) is performed to further complement the frequency information. The proposed model has been evaluated on a self-collected dataset as well as a public driving dataset. The cross-subject classification results obtained demonstrated its effectiveness. This work offers a new solution for EEG decoding, *i.e.*, using optimized RNNs for decoding complex raw EEG data and boosting the classification performance of EEG-based pBCI tasks.

1. Introduction

Electroencephalogram (EEG) is a type of fundamental neurophysiological signals that can reflect human brain activities. In our life, humans may suffer from negative mental states such as fatigue, stress and distraction. The brain-computer interface (BCI) system (Wolpaw, Birbaumer, McFarland, Pfurtscheller, & Vaughan, 2002), which can passively decode the mental states of the user by using spontaneously generated neurophysiological signals, is called the passive BCI (pBCI) (Aricò et al., 2017; Aricò, Borghini, Flumeri, Sciaraffa, & Babiloni, 2018). Among different types of neurophysiological signals, the EEG-based pBCI can be exploited to monitor specific brain activities and predict human mental states such as fatigue. Based on that, the system can then perform a series of prompts to alert the operators to take action before accidents occur. Therefore, the EEG-based pBCI can be used to predict potentially dangerous situations in dynamic and safety-critical systems such as air traffic control, aviation and self-driving cars. For instance,

in the context of air traffic control (Aricò et al., 2016), pBCI could be exploited to monitor the operators' mental workload and decide the appropriate time to enable adaptive automation (Rouse, 1976). For self-driving cars, pBCI can enable the system to detect the drivers' mental states in real time, which offers a good estimate of the driver's readiness during the pending takeover in case of a complicated road environment or traffic situation. Based on the driver's mental state, an adaptive notification can be provided to the driver (Zander et al., 2017). As such, to ensure the accurate measurement of mental states, developing effective and efficient EEG decoding methods is becoming one of the most concerning problems in EEG-based pBCI.

This work focuses on the classification aspects of EEG decoding. Various hand-engineering features have been investigated for the EEG-based pBCI tasks (Martínez Beltrán, Quiles Pérez, López Bernal, Martínez Pérez, & Huertas Celdrán, 2022; Rahman, Sarkar, Hossain, & Moni, 2022; Zheng, Yin, Wang, & Zhang, 2023). By applying the

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classification models, the extracted features can be decoded to determine the specific mental states (Duan, Zhu, & Lu, 2013). However, the hand-engineering feature extraction may discard some useful information, limiting the decoding performance. Therefore, several attempts were made to explore the possibility of decoding raw EEG data directly (Wang, Zhang, & Yao, 2021). With the development of deep learning, automatically learning from raw EEG data using deep neural networks (DNNs) (Cui, Lan, Liu, et al., 2022a; Sakalle, Tomar, Bhardwaj, Acharya, & Bhardwaj, 2021) has been recognized as a promising method for EEG decoding. Convolutional neural networks (CNNs) are one of the most popular deep learning methods for EEG processing. Different types of spatial convolution and temporal convolution have been involved in CNN models for different applications such as emotion recognition (Ding, Robinson, Zhang, Zeng, & Guan, 2022) and driver drowsiness detection (Cui, Lan, Sourina, & Müller-Wittig, 2022b). However, most of the DNNs which heavily rely on back-propagation may suffer from the issue of convergence. Specifically, the training progress may stop at a local minimum, resulting in a sub-optimal solution (Hu, Chion, Suganthan, & Katuwal, 2022; Zhang & Suganthan, 2016).

In recent years, randomized neural networks (RNNs), on the other hand, can remedy the local minima issue encountered in BP-based DNNs. RNNs create a high-dimensional space in hidden layers. This can cover the entire input space and help to reduce the risk of getting stuck in a local minimum. Moreover, the randomly initialized hidden weights are data-independent, which also ensures that the model does not suffer from the problem of overfitting. Recently, RNNs have achieved outstanding performance in different fields such as electricity load forecasting (Gao, Du, Suganthan, Zhou, & Yuen, 2022) and classification of Alzheimer's disease (Tanveer et al., 2022). The advantage of having fewer training parameters and closed-form solutions makes the RNNs suitable for the analysis of small datasets which are commonly encountered in EEG processing. Several attempts have been made to perform pBCI classification tasks by utilizing RNNs that rely on hand-engineered features, e.g., using the broad learning system (Chen & Liu, 2018) for emotion classification (Issa, Peng, & You, 2021) and using the extreme learning machine (Huang, Zhu, & Siew, 2006) for mental fatigue recognition (Yin & Zhang, 2018).

As mentioned earlier, there is a desire to decode EEG data directly from raw data. However, the use of RNNs to achieve this goal for EEG feature classification is still an area of research that needs to be explored. Three main challenges are summarized. Firstly, raw EEG data usually have high-dimension features, which may lead to the overfitting problem in the output layers of RNNs. Secondly, EEG data have the disadvantages of being non-stationary and having a low signal-to-noise ratio (SNR). Thus, there will only be a limited number of high-quality features that can be used for the classification from the randomly generated feature set. More importantly, because of the randomly initialized hidden weights, the low feature learning capability of RNNs further hinders the extraction of distinctive features and the reduction of feature dimensions. Thirdly, for cross-subject EEG decoding, the problem of subject variability (Li, Wang, & Sourina, 2022a) could limit the final classification performance.

In this work, ensemble deep random vector functional link (edRVFL) (Malik, Gao, Ganaie, Tanveer, & Suganthan, 2022b; Shi, Katuwal, Suganthan, & Tanveer, 2021) was employed as the backbone RNN for the EEG-based pBCI classification tasks due to its simple architecture and superior performance in different domains (Gao, Li, Hu, Suganthan, & Yuen, 2023). The above issues of existing methods motivated us to develop an advanced edRVFL with a powerful ability to extract valuable information from high-dimensional, non-stationary and noisy raw EEG data. In detail, to perform feature dimension reduction and automatically extract beneficial features, the canonical enhancement layer was replaced with a feature refining (FR) block to obtain a compressed representation. A subject-specific standardization strategy was used in the FR block to reduce the impact of subject variability. Then, the multiple FR blocks were stacked to enforce ensemble deep

learning. Furthermore, considering the over-high dimension and intrinsic nature of EEG data mentioned above, a novel direct link connected to the output layers named dynamic direct link (DDL) was proposed, in which non-linear features were utilized to dynamically complement the features obtained from FR blocks. To validate the effectiveness and accuracy of the proposed SedRVFL, it was evaluated on two benchmark datasets for cross-subject EEG-based pBCI tasks: stress prediction and situational awareness (SA) recognition.

In summary, this work contributes to the literature from the following perspectives:

- To the authors' best knowledge, this paper is the pioneering study that utilizes a RNN directly on high-dimension raw EEG data for classifications, unlike the classical RNNs that have been working on low-dimension hand-engineering features. Furthermore, the proposed methods can be applied to any type of RNN.
- Different from the model-based feature pruning techniques (Shi, Hu, Suganthan, & Katuwal, 2022) which may suffer from overfitting when dealing with super-high-dimension input data, a FR block that involves subject-specific standardization and unsupervised feature reduction was proposed. This was to improve the feature learning capability of the edRVFL network for decoding raw EEG data in a cross-subject setting.
- Instead of performing the propagation of linear features in the direct link which may have a negative impact on classification due to the intrinsic characteristics of raw EEG data, this work employed the propagation of non-linear features in the direct link. Not only that, to avoid overlapping of the hidden features and the non-linear direct link features, a dynamic direct link (DDL) was proposed to perform conditional propagation of direct link features to each output layer.

In the following parts of the paper, the related works are reviewed in Section 2. The details of the proposed SedRVFL are introduced in Section 3. The experiment results and the analysis of the proposed SedRVFL are presented in Section 4. The results and the advantages of the proposed model are further discussed in Section 5. Finally, the conclusion and the future work are presented in Section 6.

2. Related works

This section presents the recent advances in EEG-based pBCI classification and the related models of this work, namely edRVFL and its variants.

2.1. Passive brain-computer interface (pBCI)

BCI can be categorized into active BCI, reactive BCI and pBCI (Zander & Kothe, 2011). Different from the active BCIs such as motor imagery (MI) which relies on the user intentionally imagining specific movements and reactive BCIs such as steady-state visually evoked potential (SSVEP) in which a specific stimulus is used to evoke subjects' brain activity, the pBCI does not require any active engagement or reaction to specific stimulus from the user to generate the target brain signals. In pBCI, the target brain signals and the corresponding mental states are spontaneously produced or induced by specially designed experiments. Taking stress and SA as examples, stress can be defined as a physiological and psychological response to a perceived threat or demand that challenges an individual's ability to cope (Selye, 1956). Previous studies usually employed psychological tests such as the Stroop test to induce stress (Troyer, Leach, & Strauss, 2006). SA is defined as the perception of the elements in the environment within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future (Endsley, 1995). Considering fatigue as a low-level state of SA, previous studies have used driving experiments to induce and assess fatigue-related SA (Li, Wang, Suganthan, & Sourina, 2022b). The subjects could be of low SA because of the fatigue induced by the prolonged driving time.

2.2. Recent advances in EEG-based pBCI

Two common strategies that researchers have been using previously for EEG-based classification tasks are (1) machine learning models based on feature engineering and (2) end-to-end deep learning methods. Specifically, the first strategy exploits the hand-engineering features and performs classifier training, which is the same for all three categories of BCIs. Here, the trending hand-engineering features used in different categories of BCIs are reviewed. In the active BCIs and reactive BCIs, some of the representative features are common spatial filter patterns for MI classification (active BCI) (Ang, Chin, Zhang, & Guan, 2008) and canonical correlation analysis for SSVEP detection (reactive BCI) (Lin, Zhang, Wu, & Gao, 2006). For the EEG-based pBCI, spectral features (e.g., power spectral density (PSD) (Gu et al., 2022) and differential entropy (DE) (Zhang, Wei, Zou, & Fu, 2020) are usually involved. Entropy-based features (e.g., sample entropy (Sharma et al., 2022) and dynamic sample entropy (Lu et al., 2020)) were also investigated for mental state recognition. Specifically, in the EEG-based pBCI such as stress prediction and fatigue detection, PSD features were extensively employed. For instance, Wen and Mohd Aris (2022) extracted PSD features of Theta, Alpha and Beta frequency bands for multi-level stress classification. Ye, Yin, Zhao, Tian, and Sun (2022) identified fatigue state by exploiting PSD features. Furthermore, Gao et al. (2020) exploited the PSD and DE features to characterize the traits related to driver fatigue states. For SA recognition, PSD features were also widely employed to distinguish high SA and low SA states (Kästle, Anvari, Krol, & Wurdemann, 2021).

Compared with the hand-engineering-based classification models that exploited the features with limited information, the end-to-end DNNs have been recognized as a promising method for EEG-based classification tasks due to the better feature learning capability on raw EEG data. Previously, there have been various initiatives to investigate effective DNNs for EEG processing. In order to classify raw EEG data, Schirmer et al. (2017) proposed two different CNN architectures, the shallow version and the deep version, which outperformed the model that relied on the filter-bank common spatial pattern (FBCSP). In 2017, Lawhern et al. (2018) proposed a compact CNN called EEGNet which employed depthwise and pointwise convolution layers. The EEGNet was evaluated on different BCI paradigms. Comparable results were obtained in comparison with deep CNN and shallow CNN (Schirmer et al., 2017) in both subject-dependent and subject-independent settings. Shallow CNN, deep CNN (Schirmer et al., 2017) and EEGNet (Lawhern et al., 2018) have become the most commonly used baselines in EEG-related classification works. Narrowing down to the EEG-based pBCI, several end-to-end deep learning models have been developed. For instance, to learn the temporal dynamics and spatial asymmetry of EEG, Ding et al. (2022) presented a multi-scale CNN for emotion recognition. Khare and Bajaj (2021) also exploited a CNN to learn from two inputs which were (1) the converted time-frequency-domain data and (2) the images that were transformed from the time-domain EEG signals by using Smoothed pseudo-Wigner-Ville distribution. Furthermore, Cui, Lan, Liu, et al. (2022a) employed a compact CNN to perform cross-subject fatigue recognition from single-channel EEG. Following that, an InterpretableCNN (ICNN) for drowsiness detection (Cui, Lan, Sourina, & Müller-Wittig, 2022b) was then proposed by them. This model first performed spatial convolution to learn the cortical source and then used separable convolution to learn the temporal information. ICNN performed better at classifying drowsiness than the baselines such as EEGNet (Lawhern et al., 2018).

2.3. Ensemble deep random vector functional link (edRVFL) and its variants

The edRVFL network (Shi et al., 2021) is an extension of the RVFL (Pao & Takefuji, 1992) with a shallow structure. To achieve deep representation learning, multiple hidden layers are stacked to create the deep RVFL. Under this hierarchical structure, the input features

are fed into each hidden layer to guide the generation of a diverse set of hidden features. After that, the concatenation of the input features and the hidden features is performed and the enhancement features are then obtained. Specifically, let the input be $X \in \mathbb{R}^{K \times E}$, where K represents the number of samples and E represents the dimension of the input feature, and let the number of hidden layers be L and the hidden features of l th layer be $H^l \in \mathbb{R}^{K \times N^l}$ where N^l is the number of hidden layer nodes, the hidden features can be calculated by

$$H^1 = \phi(W^1 X), \quad (1)$$

$$H^l = \phi(W^l [H^{l-1} X]), \quad (2)$$

where W^l represents the weight matrix in the l th layer. Then, Ridge regression is employed to compute the output weights β^l . Moreover, ensemble learning is exploited to integrate the outputs o^l of classifiers from different output layers. The majority voting or averaging can be exploited to perform ensemble learning. The obtained model is edRVFL. Unlike the popular deep learning models with a single output layer, the edRVFL network trains multiple output layers based on all the hidden features. The specific architecture is presented in Fig. 1.

Several attempts have been made to improve the RVFL network (Malik, Ganaie, Tanveer, Suganthan, & Alzheimer's Disease Neuroimaging Initiative Initiative, 2022a) and the edRVFL network (Hu & Suganthan, 2022). For instance, there is an issue with edRVFL where the randomized features impede its learning ability. Thus, a model-based pruning scheme was proposed to filter out this inferior information (Shi et al., 2022). In addition, a weighting algorithm was proposed to assign different weights to the samples based on whether the samples were correctly classified in the shallow layers. Finally, the weighting and pruning-based edRVFL (WPedRVFL) was proposed (Shi et al., 2022).

In this work, for decoding raw EEG data, the reduction of feature dimension is essential. Although model-based feature selection methods have been attempted in the edRVFL network to improve the performance and reduce the dimension of hidden features (Shi et al., 2022), it is still infeasible in raw EEG decoding tasks since the over-high feature dimensions of EEG data and the limited number of samples may lead to the problem of overfitting in the model that is used to calculate the feature importance. Considering this, the use of unsupervised feature dimensionality reduction methods that can perform feature pruning and mapping to get beneficial features could be a better choice. Some attempts have been made to perform unsupervised feature learning such as deep projection in echo state network (Ma, Shen, & Cottrell, 2020). However, the issue of subject variability in EEG decoding needs to be considered in feature learning as it may limit the classification capability of the extracted features. Therefore, further research is needed to address the challenges of decoding high-dimensional raw EEG data using edRVFL.

Furthermore, in the ordinary edRVFL network (Shi et al., 2021) and its variants (Li, Gao, Yuan, Suganthan, Wang, & Sourina, 2023b; Shi et al., 2022), the direct link was leveraged to propagate linear features from the input space to the output layers to facilitate model learning. However, the propagated linear features may be less effective for EEG data due to their over-high dimensionality, non-stationary nature and low SNR characteristics. As a result, the contribution of linear patterns to the classification task may be limited. Considering this, non-linear features could be more applicable for EEG processing using the edRVFL. In this case, the propagation of the non-linear features for each output layer should be conditional to avoid overlapping between the existing random hidden features and the non-linear direct link feature.

The designed SedRVFL model, motivated by the two considerations mentioned above, is introduced in the next Section.

3. Methodology

In this section, the details of the proposed SedRVFL network are introduced. The architecture of the SedRVFL network is shown in Fig. 2.

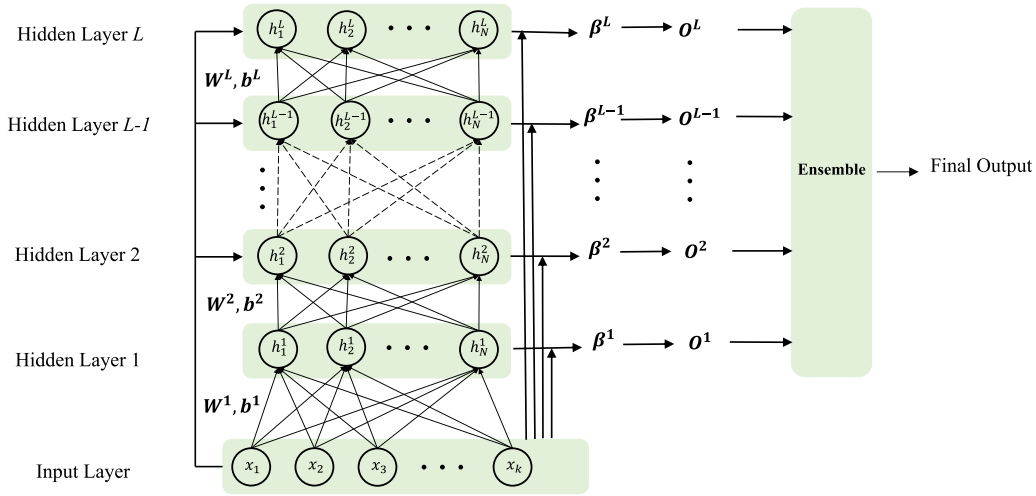


Fig. 1. Architecture of the edRVFL. x_k represents the k th input data. h_n^l represents the n th hidden neuron in the l th hidden layer. b^l represents the randomly initialized bias in l th hidden layer.

In the proposed SedRVFL, the FR blocks which consisted of a series of unsupervised learning strategies to reduce feature dimension and extract beneficial hidden features were used to replace the original hidden layers in the edRVFL network. The direct link connected to the output layers was replaced with the DDL which dynamically propagated the non-linear features and removed the redundant information. In this section, the FR block is presented first, followed by the description of the DDLs of the output layers.

3.1. Feature-refining (FR) block

The time-domain EEG data is denoted as $\mathbf{X} \in \mathbb{R}^{K \times C \times T}$, where C represents the number of EEG channels and T represents the number of time steps. In this work, the time-domain EEG data was transformed to the frequency domain. The magnitude of the transformed frequency-domain data $\mathbf{X}_f \in \mathbb{R}^{K \times C \times U}$ was then flattened and normalized to $[-1, 1]$. Here, U represents the number of frequency points after transformation. Following that, the flattened frequency domain data $\mathbf{X}_{ff} \in \mathbb{R}^{K \times d}$ was employed as the input of the SedRVFL network, where $d = C \times U$.

There were four steps involved in the FR block: (1) Hidden features calculation; (2) Low-variance feature pruning (LVFP); (3) Subject-specific standardization; and (4) Dimensionality reduction. The four steps are introduced in detail in the following parts.

For the hidden layers of edRVFL, previous studies typically set the number of hidden nodes as a hyper-parameter and optimize it at the hundreds level. In this study, after transforming to the frequency domain, the transformed EEG data included two dimensions, *i.e.*, channels and amplitudes of frequency points. As a result, the input data were typical of a super-high dimensionality, which was at the thousands level after flattening.

Hidden feature calculation. Considering the impact of high-level noises in EEG data, a low number of hidden nodes may limit the quality of the generated features for classification. Therefore, this work set a relatively large number of hidden nodes, which could increase the possibility of generating beneficial features. The feature dimension of the calculated hidden feature $\mathbf{H}$ is denoted as N_b .

Low-variance feature pruning (LVFP). Among the obtained hidden features in the specific hidden layers, the desired features should contain adequate information rather than similar content in the samples from different classes. In this step, less beneficial features were discarded to reduce their negative impact on the classification performance. The variance was employed to first evaluate the amount

of information contained in the hidden features. Then, the features with limited or zero information for classification were discarded. Specifically, the index for feature pruning and selection (PSI) of the hidden features can be obtained by

$$PSI = \begin{cases} 0, & Var(i) < \alpha \\ 1, & Var(i) \geq \alpha \end{cases} \quad (3)$$

where α is the threshold for the variance of the hidden features and i is the index of the hidden features. If $PSI_i = 0$, the i th hidden feature was selected. Otherwise, the i th hidden feature was pruned. Then, by applying PSI to the hidden features $\mathbf{H}$, the selected features $\mathbf{H}_v$ can be calculated by

$$\mathbf{H}_v = [\mathbf{H}_i]_{i: PSI_i > 0}, \quad (4)$$

As shown in Fig. 2, the selected features obtained from LVFP were then concatenated with the frequency-domain input propagated by the ordinary direct link, and then fed into the next FR block to guide the generation of the hidden features in the next layer. In detail, to propagate the integral hidden features to the next FR block, the high-variance features $\mathbf{H}_v$ are concatenated with the input frequency-domain data $\mathbf{X}_{ff}$ by $\mathbf{H}_i = [\mathbf{X}_{ff} \mathbf{H}_v]$. Then, the obtained enhancement features are inputted into the next FR block.

Subject-specific standardization. To deal with the problem of subject variability, which remains a challenge in existing RNN models, this study took inspiration from a previous study (Li, Gao, & Suganthan, 2023; Li, Wang, & Sourina, 2022a) which demonstrated the beneficial effect of subject-specific (domain-specific) batch normalization statistics in reducing subject variability. The present work introduced a novel approach of subject-specific standardization to tackle the issue of subject variability. Specifically, the selected features $\mathbf{H}_s$ of each subject were standardized and concatenated to obtain a set of standardized features as given by

$$\mathbf{H}_s = \left[\frac{\mathbf{H}_v^1 - \mu^1}{\sigma^1}, \dots, \frac{\mathbf{H}_v^j - \mu^j}{\sigma^j} \right]_{1 \leq j \leq M}, \quad (5)$$

where $\mathbf{H}_v^j$ represents the selected features of subject j . μ^j and σ^j denote the mean and standard deviation (STD) of $\mathbf{H}_v^j$, respectively. M represents the number of subjects in the dataset.

Dimensionality reduction. The final step of the FR block continued with reducing the feature dimension and further extracting features with adequate information. In the literature, there are mainly two ways for dimensionality reduction, namely supervised learning methods and unsupervised learning methods. Although the LVFP has been performed

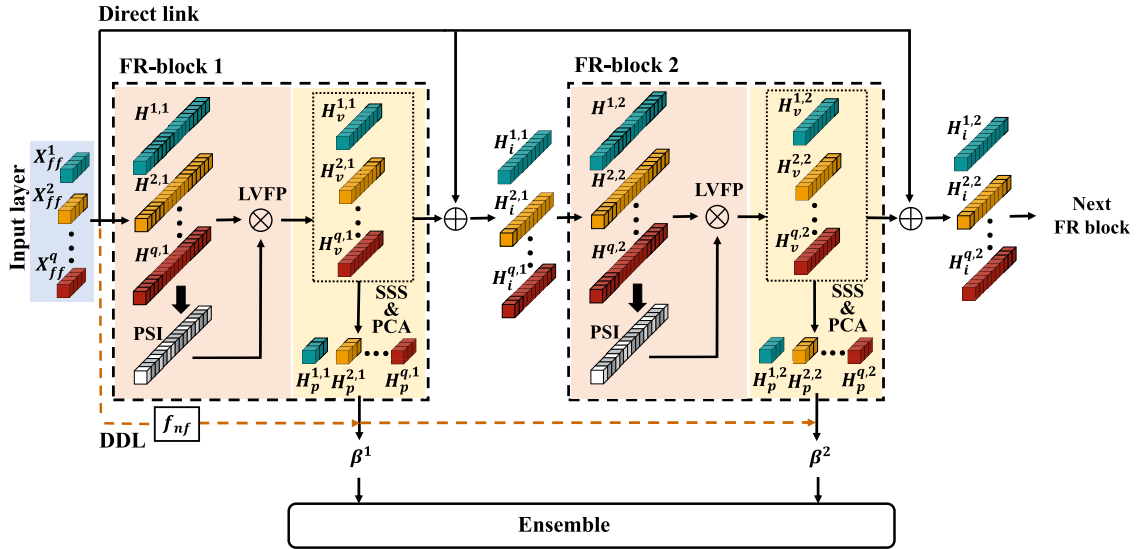


Fig. 2. Architecture of the SedRVFL network. The Gray cubes represent that the corresponding features are with high variance, while the white cubes denote that the corresponding features are with low variance. $\otimes$ is the element-wise product and $\oplus$ means the concatenation along the feature dimension. SSS is subject-specific standardization. X_{ff}^j represents the flattened frequency-domain data of subject j , where $j \in \{1, 2, \dots, q\}$. $H^{j,l}$ means the hidden feature of subject j in l th layer, which is same for $H_v^{j,l}$, $H_p^{j,l}$ and $H_i^{j,l}$. f_{nf} represents the non-linear mapping in the DDLs of output layers.

before this step, the resulted hidden features may still have a super-high dimension, which may lead to the overfitting problem for the models used for supervised learning. Therefore, further feature dimensionality reduction and feature refining based on unsupervised learning methods are desired. PCA is utilized in this work. It is a representative unsupervised dimensionality reduction method that can extract features that are potentially more distinctive by projecting the data onto the identified directions with higher variance. The comparison between PCA and other supervised and unsupervised dimensionality reduction methods was carried out, as described in Section 4.5.

PCA can be used to map the input to the components with the variance from large to small, i.e., from high significance to low significance. In mathematics, the matrix that consists of eigenvectors A can be obtained by calculating the eigenvalues of the feature's covariance matrix C_{xx} , which can be represented as

$$AC_{xx}A^T = \begin{Bmatrix} a_1 & & \\ & \ddots & \\ & & a_B \end{Bmatrix}, \quad (6)$$

where $C_{xx} = H_s^T H_s$. B is the number of features in H_s . From a_1 to a_B , the significance of the corresponding eigenvectors decreases from high to low. Based on the requirements, the matrix that consisted of the first b eigenvectors A_b with higher significance was selected. Then, the function of the feature refining step can be defined by

$$H_p = f_{ref}(H_s) = H_s A_b, \quad (7)$$

where H_p represents the feature obtained after the dimensionality reduction by PCA.

Overall, the FR block performed a series of unsupervised feature extraction to reduce feature dimensionality, lessen the impact of noises and subject variability and obtain beneficial features for better classification performance. Worth mentioning, the subject-specific standardization step provided a good initialization for PCA, allowing the extracted features to be effective for unseen subjects.

3.2. Dynamic direct link (DDL)

In the classical RVFL and edRVFL networks, the direct link benefits model learning in two aspects: (1) guiding the generation of hidden features of different hidden layers and (2) propagating the

linear features from the input layer to the output layers. In this work, the ordinary direct link was adopted to execute the first mentioned function. For the second function of direct link, two changes were proposed in consideration of the intrinsic characteristics of EEG and the characteristics of the proposed SedRVFL network. Firstly, due to the EEG characteristics of non-stationary and having low SNR, fewer linear patterns from the input signals could be of real value for classification. Furthermore, the over-high dimension of input data may lead to the problem of overfitting for the classification model in the output layers. Therefore, this work considered converting the input to non-linear features and propagating them to the output layer. By denoting the function of conversion to non-linear features as $f_{nf}(\cdot)$, the features to be propagated can be obtained by $f_{nf}(X_{ff})$. Then, the enhancement features obtained at the output layer can be denoted as $D_f = [f_{nf}(X_{ff}) H_p]$.

Based on the propagation of non-linear features, the second change was proposed. Specifically, since the refined hidden features were generated from PCA of the FR block by non-linear mapping as well, it could be possible to encounter extensive overlapping between the refined features and the features propagated from the direct link, resulting in performance deterioration. To deal with this problem, the DDL was proposed, which employed a discriminant hyper-parameter to dynamically combine the non-linear features and the refined hidden features. Specifically, the decision of propagation was based on the correlation matrix ρ between the non-linear features from the direct link and the refined features of the highest significance. Thus, the enhancement feature obtained in l th output layer can be represented as

$$D_f = \begin{cases} H_p, & \theta < c < 1 \\ [f_{nf}(X_{ff}) H_p], & 0 < c < \theta \end{cases} \quad (8)$$

where $c = \text{mean}(\rho)$ and θ is the discriminant.

By applying the DDL, the classifier of the l th output layer can be solved by

$$\text{prime space: } \beta^l = ((D_f^l)^T D_f^l + \lambda^l I)^{-1} (D_f^l)^T Y, \quad (9)$$

$$\text{dual space: } \beta^l = (D_f^l)^T (D_f^l (D_f^l)^T + \lambda^l I)^{-1} Y, \quad (10)$$

where λ is the regularization parameter.

3.3. Spectral ensemble deep random vector functional link (SedRVFL)

The algorithm of the SedRVFL that involved the FR blocks and the DDL algorithm is presented in Algorithm 1. In the proposed SedRVFL, the time-domain data were first transformed to the frequency domain. The magnitude component of the frequency-domain data was input to the SedRVFL network. During training, the hidden features were first calculated based on the flattened frequency-domain data. Then, the LVFP was performed to select features with adequate information by using variance. The selected features were combined with the frequency-domain input propagated by the ordinary direct link, and then fed into the next FR block to guide the generation of the hidden features in the next layer. At the same time, the selected features of each subject were standardized individually and fed into PCA to obtain the refined features by non-linear mapping. The refined features were then fed into the output layers. In each output layer, the DDL algorithm was performed. The input was first converted to non-linear features and propagated dynamically to each output layer based on the discriminant hyper-parameter to complement the refined hidden features. The obtained enhancement features were used to calculate the weights of the output layers. During testing, the same process used in training was adopted to calculate the enhancement features which were then utilized to compute the final outputs. Following that, the outputs from all the output layers were integrated by averaging to obtain the final outputs.

Algorithm 1: Training algorithm for the SedRVFL

During training:

Input: $X_{train} \in \mathbb{R}^{K \times C \times T}$

Output: Output layer parameters, β^l

Perform Fourier transform on the X_{train} and obtain $\mathcal{F}(X_{train})$

Flatten $\mathcal{F}(X_{train})$ as X_{ff}

Initialize the hidden weights W^l and bias b^l randomly

$l = 1$

for $l \leq L$ **do**

 Calculate the hidden features by using Eqs. (1) and (2)

 Select the features with high variance, H_v , by using Eqs. (3) and (4)

 Perform subject-specific standardization by using Eq. (5)

 Refine features by using Eqs. (6) and (7)

 Perform the DDL by using Eq. (8)

 Calculate the output weight β^l by using Eqs. (9) and (10)

 Generate the input of the next layer which is $H_l = [X_{ff} \ H_v]$

$l++$

4. Experiments

In this section, the proposed SedRVFL network was evaluated on two EEG-based pBCI classification tasks, including stress prediction and SA recognition. The dataset introduction, experiment settings and the evaluating results are introduced and discussed in the following subsections.

4.1. Dataset introduction

4.1.1. Self-collected stress dataset

Ten participants (nine males and one female), aged between 25 and 33 years ($M = 25.42$, $SD = 3.66$), were involved. The study was conducted among internal staff and students as part of the experiment design process. All the participants had normal or corrected to normal visual acuity. None of them were suffering from mental disorders, chronic and cardiovascular diseases, or under any medical treatment in the last 6 months.

The overall experiment procedure is shown in Fig. 3. The Stroop test (Troyer et al., 2006) was used to induce the stress states of the subjects and it was programmed by using the Psychology Experiment Building Language (PEBL) (Mueller & Piper, 2014). There were two tasks in the Stroop test, the congruent task and the incongruent task. In

the congruent task, the participants were presented with color-words printed in a consistent color ink (e.g. the word “green” is printed in green ink). In the incongruent task, the participants were presented with color-words in conflicted ink (e.g. the word “green” is printed in blue ink). Examples of these two tasks are shown in Figs. 4 and 5.

In both tasks, the participants were tasked with naming the ink of color-words by reading the references provided at the bottom of the screen and pressing the corresponding keys on the keyboards as quickly as possible. As naming the ink color was a less automated task than reading the name of the color-word, the difficulty in inhibiting the more automated process was created (Stroop, 1935) and this has been widely used to induce stress (Hou et al., 2015; Kurniawan, Maslov, & Pechenizkiy, 2013). Different from the classical Stroop test, the corresponding keys of the reference were shuffled after each trial in this work. This was to mitigate the impact of memorization on the experiment.

Based on the described Stroop test, the experiment was divided into two sessions: the practice session and the test session. The practice session consisted of a congruent task with eight trials and an incongruent task with eight trials to help the participants familiarize themselves with the tasks and further reduce learning effects. Here, a trial means a query of a color word in the Stroop test. Examples of the congruent task and the incongruent task of the practice session are shown in Fig. 4.

In the test session, four blocks were designed, which aimed to induce different stress levels. Each block lasted for 5 minutes, during which the page consisting of 24 trials was presented in a loop. The trials in all blocks were randomly generated. Blocks 1 and 2 were designed to induce low levels of stress by using the congruent task, while blocks 3 and 4 were expected to induce high levels of stress by incongruent color words. Examples of the congruent task and the incongruent task in the test session are shown in Fig. 5. Specifically, in block 1, the participants were required to watch a trial demonstration video on the congruent task of the designed Stroop test, in which the queries were automatically answered. This was to collect the resting state data of the subjects. In block 2, a congruent task was executed. In block 3, an incongruent task was performed. In block 4, an incongruent task was performed. While the total duration of the block remained as 5 minutes, the participants were given a time limit to respond for each trial. The duration of the time limit was the average of the participant’s response time for each trial in the incongruent task of the practice session. If no response was given within this time limit, it would be considered a false answer. The next trial would continue.

The effectiveness of the tasks in inducing different stress levels was evaluated by the short stress state questionnaire (SSSQ) (Helton & Näswall, 2015) answered by the participants. The SSSQ is a self-report questionnaire used to assess the stress levels of the participants. The questionnaire consisted of 24 items that assess the different aspects of stress, including Distress, Worry, and Engagement. Distress appears to integrate unpleasant mood and tension with the lack of confidence and perceived control (Helton, 2004). Worry is a cognitive factor primarily composed of self-focused attention, self-esteem, and cognitive interference (Helton, 2004). Task Engagement refers to the integration of energetic arousal, motivation, and concentration constructs that are related to task interest and focus (Helton, 2004). The participants were asked to rate each item on a 5-point Likert scale, indicating the extent to which they had experienced each symptom of stress before the test session of the experiment using the State Pre-Questionnaire of SSSQ. They were also needed to complete the State Post-Questionnaire of SSSQ after each block of the test session as shown by Fig. 3. The used SSSQ is presented in Supplementary Fig. A.1. The average self-rating scores of the three primary symptoms of stress (Distress, Worry, and Engagement) in the SSSQ questionnaire (Helton & Näswall, 2015) at different stages of the experiment were shown in Table 1. It was observed that the overall stress level of the participants increased along with the execution of the experiment (the distress and worry level

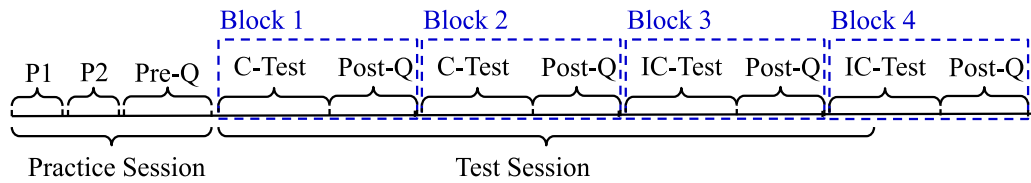
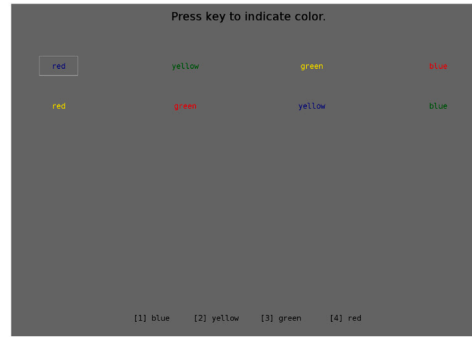
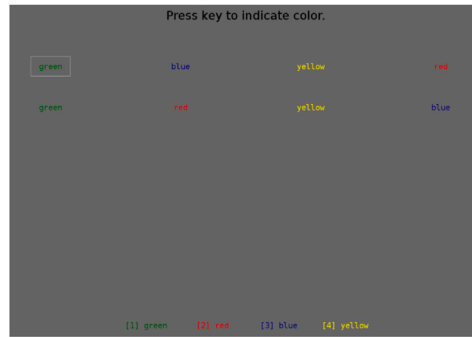
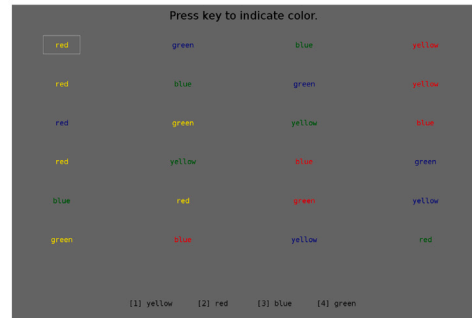


Fig. 3. Experimental procedure. P1 and P2 represent the first and second practice session, respectively. C-Test means the congruent task of the Stroop test. IC-Test represents the incongruent task of the Stroop test. Pre-Q represents the State Pre-Questionnaire of SSSQ while Post-Q represents the State Post-Questionnaire of SSSQ.



(a) Practice session 1: Congruent task (b) Practice session 2: Incongruent task

Fig. 4. Congruent task and incongruent task in the practice session.



(a) Congruent task (b) Incongruent task

Fig. 5. Congruent task and incongruent task in the test session.

Table 1

The scores of three primary factors in the SSSQ questionnaire.

	Pre-questionnaire	Block 1	Block 2	Block 3	Block 4
Distress	1.20	1.36	1.40	1.41	2.26
Worry	2.35	2.25	2.05	2.07	2.20
Engagement	3.72	3.54	3.71	3.62	3.18

increased from level 2 to 4, while the engagement level decreased), which provided the reference for the following data labeling.

For an initial test, a 2-class (low stress versus high stress) classification problem was utilized to justify the capability of the SedRVFL network. Data collected from block 2 over that from block 1 as low-stress data, in order to exclude the task-discriminative features in the data collected from block 1. The data collected from block 4 was selected as the data for the high-stress class since they could contain more discriminative features related to mental stress. Therefore, the data of blocks 2 and 4 were used to perform a binary stress prediction task.

EEG and eye-tracking were used to collect data on mental states. Heart rates were collected as well. EEG signals were used for this

work. The EEG signals were recorded using a 14-channel headset (Emotiv Epoc+ Pro) following the International 10–20 system of electrode placement. The raw EEG signals were sampled at 128 Hz. In the pre-processing, the high-pass filter of 1 Hz was performed. A moving window of 10 s in length with a stride of 1 s was used to extract the data. Finally, 6150 samples were obtained. The data size of one sample was 14 (channels) $\times$ 1024 (time steps).

4.1.2. Public driving dataset

For the task of SA recognition, this study utilized a pre-processed public driving dataset (Cao, Chuang, King, & Lin, 2019) to evaluate the proposed model. In the experiment for this dataset, lane-departure events were randomly induced to make the car drift from the original

cruising lane towards the left or right side (deviation onset). Each participant was instructed to quickly compensate for this perturbation by steering the wheel (response onset) to cause the car to move back to the original cruising lane (response offset). Deviation onset, reaction onset and response offset events were all included in a complete trial.

Fatigue-related SA was the focus of our research. We extracted 3 s of the EEG data prior to the deviation onset. Then, the procedure used in the work by Cui, Lan, Liu, et al. (2022a) was followed and the local reaction time (RT) and the global RT were used to label data. When both the local and global RT were shorter than 1.5 times the alert-RT, the corresponding extracted EEG data was labeled as “high SA”. When both the local and global RT were longer than 2.5 times the alert-RT, the data was labeled as “low SA”. Transitional states with moderate performance were excluded and the neutral state was not considered in this work. When there was more than one dataset for a subject, the most balanced one was chosen to execute the filtering. Then, the data were further down-sampled to 128 Hz. Finally, a balanced SA dataset was obtained which included 2022 samples of 11 subjects. The data size of one sample was 30 (channels)×384 (time steps).

4.2. Experiment setting

The performance of the proposed SedRVFL network was evaluated on the challenging cross-subject stress prediction and SA recognition tasks. The leave-one-subject-out (LOSO) cross-validation (CV) was conducted and the proposed model was compared with 8 baseline methods, which are listed below:

- (1) Ridge regression with the input of PSD features;
- (2) Logistic regression (LR) with the input of PSD features;
- (3) Support vector machine (SVM) (Wei, Wang, Lin, & Jung, 2018) with the input of PSD features;
- (4) EEGNet (Lawhern et al., 2018) with the input of raw time-domain data;
- (5) ShallowCNN (Schirmermeister et al., 2017) using raw time-domain data;
- (6) RVFL (Pao & Takefuji, 1992) with the input of raw frequency-domain EEG data;
- (7) Extreme learning machine (ELM) (Suganthan & Katuwal, 2021) with the input of raw frequency-domain EEG data;
- (8) EdRVFL (Shi et al., 2021) with the input of raw frequency-domain EEG data.

In addition, the state-of-the-art (SOTA) method for cross-subject EEG-based classification, ICNN (Cui, Lan, Sourina, & Müller-Wittig, 2022b), and the SOTA method of the RNN, WPedRVFL (Shi et al., 2022), were included in the comparison as well. The inputs for the two SOTA methods were raw time-domain data and raw frequency-domain data, respectively. Notably, the ICNN was originally designed for the classification task on the public driving dataset used in this work and has been shown its good classification capability in different tasks (Li, Wang, Suganthan, & Sourina, 2022b).

In the proposed SedRVFL network, PSD features were used as the non-linear features in the DDL as mentioned in Section 3.2. The used PSD features in the baseline methods and the DDL of the SedRVFL network were computed from four spectral bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz) and beta (12–30 Hz). The activation function for all the RNNs was *Tanh*. Regarding the hyper-parameter setting of the back propagation-based deep learning models, Adam optimizer with momentum $\beta_1 = 0.9$ and $\beta_2 = 0.99$ was used. The mini-batch and the learning rate were 50 and 0.001, respectively. During the training, the model was trained for 70 epochs and the early stop was performed based on the LOSO CV on the training set. Cross-entropy was employed as the cost function for training. The mentioned hyper-parameters were the same for all the evaluated CNN models. For the edRVFL, WPedRVFL and SedRVFL networks, the maximum number of hidden layers was set as 10. The remaining hyper-parameters of the compared methods

Table 2

Search space of the classification methods.

Method	Hyper-parameter	Search Space
Ridge	L2 regularization	[0.001, 0.03125, 0.0625]
SVM	Regularization parameter	[2, 4, 8]
LR	Regularization parameter	[2, 4, 8]
EEGNet	F1	[4, 8, 16, 32, 64]
ShallowCNN	The number of channels	[16, 32, 64]
ICNN	N1	[16, 32, 64]
RVFL, ELM, edRVFL	The number of hidden nodes L2 regularization	[30 000, 35 000, 40 000] [0.001, 0.03125, 0.0625]
WPedRVFL	The number of hidden nodes L2 regularization Pruning rate Weighting parameter	[30 000, 35 000, 40 000] [0.001, 0.03125, 0.0625] [0.3, 0.5, 0.7, 0.9] [0.5, 0.7, 0.9]
SedRVFL	N_b α λ b θ	[30000, 35000, 40000] [0.1, 0.15] [0.001, 0.03125, 0.0625] [70, 90, 110, 130, 150] [0.16, 0.32, 0.48]

were optimized by using a grid search with the LOSO CV that was performed on the training set. The search space of the hyper-parameters was presented in Table 2.

4.3. The experimental results of stress prediction

The average accuracy, STD and median of the cross-subject stress prediction using the proposed SedRVFL network and the baseline methods were shown in Table 3. Overall, the proposed SedRVFL network showed a better average accuracy than the baseline methods. Worth noting, there was an approximately 9% improvement compared with the SOTA classification method, ICNN (Cui, Lan, Liu, et al., 2022a). Thus, the effectiveness of the SedRVFL network on the cross-subject stress prediction was demonstrated.

Moreover, in the comparison with RNNs, the poor EEG decoding capability of the compared RNNs was observed as only approximately 60% average accuracy was obtained. In contrast, the proposed SedRVFL network achieved 78.95% average accuracy. This indicated that the FR block and the DDL enabled the edRVFL to be capable of handling the challenging high-dimension data decoding.

It was observed that the accuracy for some subjects obtained by the proposed method was not as good as the baselines. In order to further compare the general performance, the median of the individual accuracy was compared. Specifically, the SedRVFL network achieved 80.21% median accuracy, which was much higher than the compared baselines. This illustrated that the proposed model could be more suitable for cross-subject classification tasks.

The one-tailed Wilcoxon paired signed-rank statistical test was conducted to compare the proposed model with other methods in a pairwise manner. The p -values were shown in Table 4. It was observed that the proposed SedRVFL network showed significant performance improvement over most of the baseline methods with $p < 0.05$.

Table 3

Average accuracy, STD and median (%) on LOSO stress prediction.

Method	Subjects										Avg. Acc.	Std.	Median
	1	2	3	4	5	6	7	8	9	10			
Ridge	49.34	51.92	81.57	61.71	87.64	56.42	76.22	64.31	87.01	51.96	66.81	15.04	63.01
EEGNet (Lawhern et al., 2018)	57.87	26.88	97.88	67.69	89.43	71.32	32.76	60.20	84.54	85.29	67.38	23.64	69.50
LR	14.10	46.41	83.52	61.07	73.98	72.90	67.09	66.78	79.44	76.31	64.16	20.48	69.99
SVM (Wei et al., 2018)	50.66	51.09	60.03	60.10	92.03	60.70	67.24	50.99	91.12	59.15	64.31	15.31	60.06
ICNN (Cui, Lan, Sourina, & Müller-Wittig, 2022b)	45.90	63.61	95.92	51.70	88.78	80.19	57.95	58.06	66.78	93.79	70.27	18.10	65.19
ShallowCNN (Schirmer et al., 2017)	64.43	56.76	94.78	49.60	86.67	69.10	35.43	81.25	47.53	94.61	68.02	20.84	66.76
ELM (Suganthan & Katuwal, 2021)	43.28	63.61	59.87	51.21	73.01	52.46	55.28	53.62	60.86	70.92	58.41	9.15	57.57
RVFL (Pao & Takefuji, 1992)	48.69	62.77	59.71	51.86	69.43	53.25	53.70	51.81	64.31	72.06	58.76	8.11	56.70
edRVFL (Shi et al., 2021)	44.26	66.61	62.32	54.60	74.15	53.88	57.32	53.95	72.37	73.53	61.30	10.15	59.82
WPedRVFL (Shi et al., 2022)	44.26	66.61	62.32	54.60	74.47	53.88	57.17	54.11	72.37	73.53	61.33	10.19	59.74
SedRVFL	69.02	81.14	95.11	57.35	82.93	76.07	60.47	79.28	92.43	95.75	78.95	13.58	80.21

Table 4Wilcoxon test results (p -value) on LOSO stress prediction.

	Ridge	EEGNet	LR	SVM	ICNN	ShallowCNN	ELM	RVFL	edRVFL	WPedRVFL
SedRVFL	0.032	0.032	0.010	0.032	0.053	0.019	0.001	0.001	0.001	0.001

Table 5

Precision, sensitivity, specificity and F1-score (%) on LOSO stress prediction.

Method	Precision	Sensitivity	Specificity	F1-score
Ridge	66.83	64.82	68.81	65.81
EEGNet (Lawhern et al., 2018)	66.66	67.43	67.31	67.04
LR	71.24	64.39	74.80	67.64
SVM (Wei et al., 2018)	64.15	62.44	66.19	63.28
ICNN (Cui, Lan, Sourina, & Müller-Wittig, 2022b)	70.78	67.46	73.01	69.08
ShallowCNN (Schirmer et al., 2017)	68.50	64.45	71.28	66.42
RVFL (Pao & Takefuji, 1992)	57.74	60.16	57.32	58.92
ELM (Suganthan & Katuwal, 2021)	57.40	59.83	56.96	58.59
edRVFL (Shi et al., 2021)	60.39	61.81	60.71	61.09
WPedRVFL (Shi et al., 2022)	60.43	61.81	60.77	61.11
SedRVFL	77.43	80.48	77.27	78.92

For further analysis, four classification metrics, namely precision, sensitivity, specificity and F1-score of the baseline methods and the proposed SedRVFL network, were compared in Table 5. The results indicated that the proposed SedRVFL network outperformed the baseline methods in all analyzed classification metrics. In particular, the best F1-score of 78.92% was achieved by the proposed SedRVFL network.

4.4. The experimental results of situational awareness (SA) recognition

To further understand the decoding capability of the proposed SedRVFL network, it was then evaluated on the public driving dataset for SA recognition. The average accuracy, STD and median of the cross-subject SA recognition task were shown in Table 6. Similar findings were obtained. Specifically, the proposed SedRVFL network achieved the highest average accuracy. In addition, the SedRVFL network substantially improved the raw frequency-domain data decoding of the original edRVFL network, achieving an increase in average accuracy of more than 7%. This could be attributed to the efficacy of the developed FR block and the DDLs. Furthermore, the SedRVFL network achieved the best median accuracy amongst the compared methods, which demonstrated its better classification capability than the compared methods. Then, the one-tailed Wilcoxon paired signed-rank was conducted with p -values shown in Table 7. The analysis showed that there was a significant performance improvement by using the proposed SedRVFL network with $p < 0.05$. The four classification metrics, including precision, sensitivity, specificity and F1-score, were compared in Table 8. It was observed that the SedRVFL network achieved the best F1-score of 79.15% among all the compared methods.

In these two classification tasks, stress prediction and SA recognition, although Wilcoxon results did not show a statistically significant difference in the accuracy as compared with the ICNN, the better classification performance of the SedRVFL network could be demonstrated in terms of precision, specificity and F1-score. Furthermore, the central

tendency measured in terms of average accuracy and median also illustrated that the SedRVFL network outperformed the ICNN. Therefore, it could be suggested that the SedRVFL network was generally better than the ICNN in these two EEG-based pBCI classification tasks.

Based on the evaluations above, it was concluded that the proposed SedRVFL network could achieve outstanding performance in the EEG-based pBCI classification tasks, namely stress prediction and SA recognition. Furthermore, the proposed SedRVFL network demonstrated better feature learning capability on high-dimension input data than the compared RNNs, achieving substantial performance improvement. Lastly, the comparison of the four classification metrics further demonstrated the superiority of the proposed method over the strong baseline methods. Therefore, the SedRVFL network can be considered a highly competitive classifier for these two EEG-based pBCI classification tasks.

4.5. Ablation study

In this section, the necessity of each component in the proposed SedRVFL network was investigated.

Firstly, the effectiveness of the components in the FR block was studied, where the DDL was not involved first. The edRVFL with the FR block is denoted as FR-edRVFL here. We compared the average accuracy of the FR-edRVFL and its variants in which the three components, subject-specific standardization, LVFP and PCA, were removed one by one for evaluation. The average accuracy of the following models was summarized in Fig. 6.

- FR-edRVFL w/o SSS: the FR-edRVFL without the subject-specific standardization.
- FR-edRVFL w/o LVFP: the FR-edRVFL without the LVFP.
- FR-edRVFL w/o PCA: the FR-edRVFL without the dimensionality reduction which exploited PCA.

Table 6

Average accuracy, STD and median (%) on LOSO SA recognition.

Methods	Subjects											Avg.	Std.	Median
	1	2	3	4	5	6	7	8	9	10	11			
Ridge	60.64	53.03	73.33	73.65	78.57	51.81	57.84	62.12	78.66	53.70	49.56	62.99	11.10	60.64
LR	45.74	84.85	73.33	70.95	68.30	77.11	67.65	70.45	80.89	67.59	53.10	69.09	11.32	70.45
SVM (Wei et al., 2018)	77.13	45.45	69.33	65.54	83.04	72.89	59.80	68.18	85.35	71.30	63.27	69.21	11.11	69.33
EEGNet (Lawhern et al., 2018)	56.38	71.97	46.67	65.54	53.57	63.86	67.65	53.79	71.66	79.63	58.41	62.65	9.82	63.86
ShallowCNN (Schirmermeister et al., 2017)	82.98	48.48	80.67	69.59	80.36	76.51	62.75	70.08	84.08	63.89	81.86	72.84	11.11	76.51
ICNN (Cui, Lan, Sourina, & Müller-Wittig, 2022b)	77.66	61.36	82.67	77.03	89.73	84.34	63.73	70.08	89.49	75.93	73.45	76.86	9.42	77.03
RVFL (Pao & Takefuji, 1992)	71.81	51.52	67.33	60.14	79.46	75.90	60.78	67.05	84.08	63.89	61.06	67.55	9.57	67.05
ELM (Suganthan & Katuwal, 2021)	72.34	50.00	66.67	59.46	79.02	75.90	60.78	67.42	84.39	63.89	61.06	67.36	9.92	66.67
edRVFL (Shi et al., 2021)	84.57	52.27	68.67	63.51	71.43	80.72	67.65	67.80	86.31	70.37	61.50	70.44	10.17	68.67
WPedRVFL (Shi et al., 2022)	84.57	55.30	67.33	62.84	72.32	78.92	67.65	68.18	86.94	70.37	62.39	70.62	9.60	68.18
SedRVFL	89.36	62.88	81.33	66.22	86.61	77.71	69.61	77.65	90.76	86.11	66.37	77.69	10.08	77.71

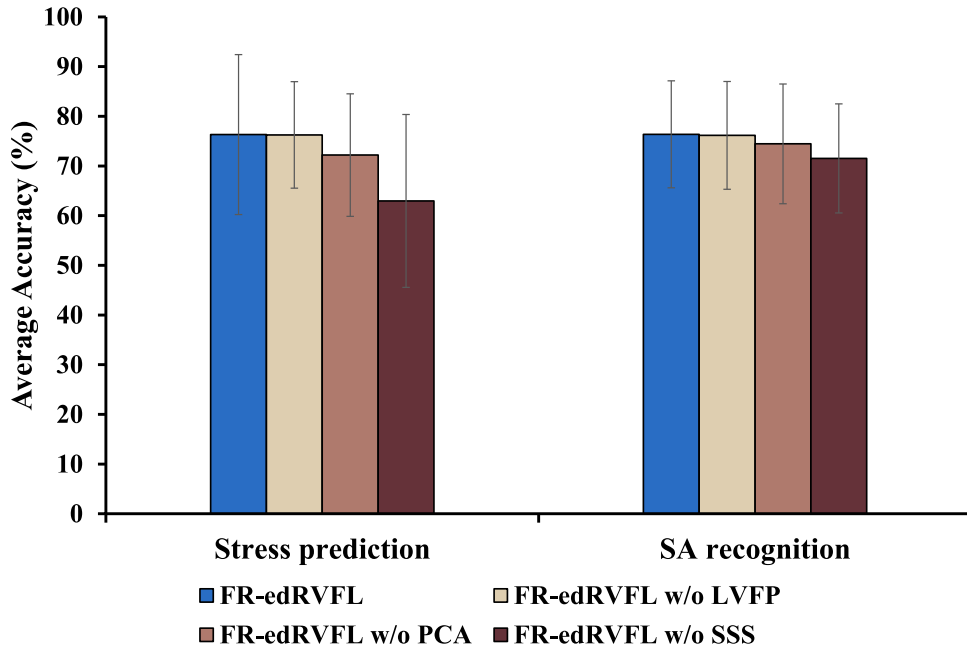
Table 7Wilcoxon test results (p -value) on LOSO SA recognition.

	Ridge	LR	SVM	EEGNet	ShallowCNN	ICNN	RVFL	ELM	edRVFL	WPedRVFL
SedRVFL	0.001	0.042	0.000	0.005	0.042	0.382	0.001	0.001	0.002	0.001

Table 8

Precision, sensitivity, specificity and F1-score (%) on LOSO SA recognition.

Methods	Precision	Sensitivity	Specificity	F1-score
Ridge	61.27	79.03	50.05	69.03
LR	72.41	61.52	76.56	66.52
SVM (Wei et al., 2018)	73.00	67.66	74.98	70.23
EEGNet (Lawhern et al., 2018)	57.59	88.92	34.52	69.91
ShallowCNN (Schirmermeister et al., 2017)	74.21	76.85	73.29	75.51
ICNN (Cui, Lan, Sourina, & Müller-Wittig, 2022b)	77.33	80.32	76.46	78.80
RVFL (Pao & Takefuji, 1992)	70.20	68.05	71.12	69.11
ELM (Suganthan & Katuwal, 2021)	70.06	68.05	70.92	69.04
edRVFL (Shi et al., 2021)	72.74	69.93	73.79	71.31
WPedRVFL (Shi et al., 2022)	73.07	70.03	74.18	71.52
SedRVFL	78.69	79.62	78.44	79.15

**Fig. 6.** Average accuracy of edRVFL with the FR block (FR-edRVFL) and its variants with different components of the FR block removed.

- FR-edRVFL (with subject-specific standardization, LVFP, PCA).

Fig. 6 demonstrated that all components of the FR block contributed positively to the feature learning of the raw frequency-domain EEG data in this work. In particular, the average accuracy deteriorated to the largest extent when subject-specific standardization was removed

(FR-edRVFL w/o SSS), demonstrating its effectiveness in both cross-subject classification tasks. A decrease in performance was also observed when PCA or LVFP were removed, but to a smaller extent as compared to removing subject-specific standardization (FR-edRVFL w/o PCA, FR-edRVFL w/o LVFP). The outstanding feature refining capability of PCA was further indicated when compared with a

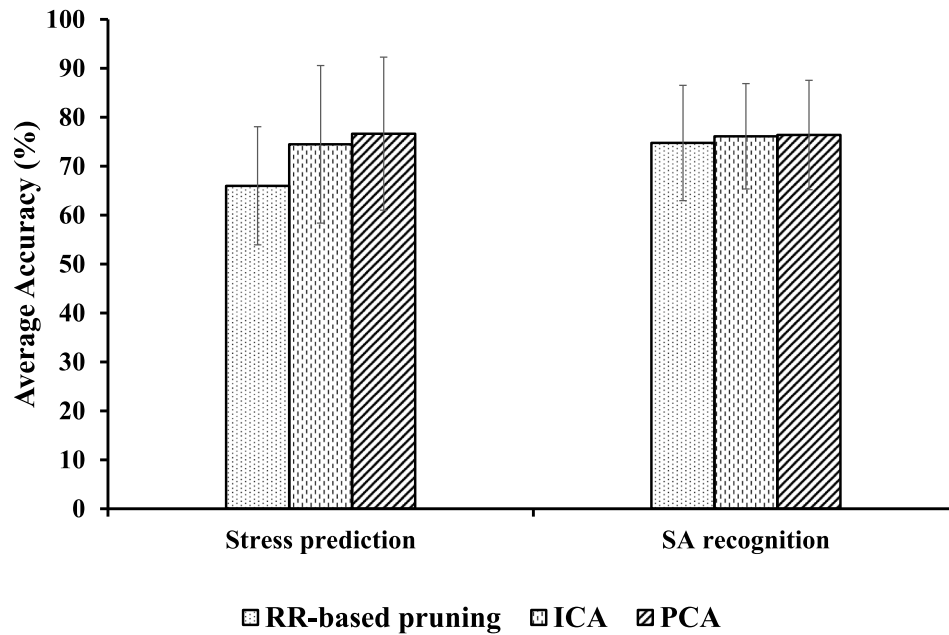


Fig. 7. Average accuracy obtained by applying different feature refining methods in the FR-edRVFL. RR represents Ridge regression.

representative supervised dimensionality reduction method: Ridge regression-based pruning technique (Shi et al., 2022) and a representative supervised dimensionality reduction method: independent component analysis (ICA) (Comon, 1994). In the hyper-parameter setting of these two methods, the search space of the number of dimensions obtained was the same as that of the number of eigenvectors selected b in PCA. The search space of the L2 regularization parameter was the same as that of the L2 regularization used in WPedRVFL as shown in Table 2. The grid search was also used to optimize the maximum iteration parameter of ICA in the search space [50, 200, 500]. The comparison results are shown in Fig. 7, with the remaining components of SedRVFL fixed, the average accuracy obtained by the FR-edRVFL with PCA was higher than the FR-edRVFL with the Ridge regression-based pruning or ICA in both classification tasks. Overall, based on the analysis, the effectiveness of the components in the FR block was demonstrated.

On top of the FR-edRVFL, the effectiveness of the DDL and the proposed propagation of non-linear features instead of linear features were analyzed. Specifically, the average accuracy of the following models were compared and the results are shown in Fig. 8.

- Raw-edRVFL: edRVFL with the input of raw frequency-domain EEG data.
- FR-edRVFL with the input of raw frequency-domain EEG data.
- LDL-FR-edRVFL: the FR-edRVFL with the ordinary direct link propagating the flattened frequency-domain EEG data to the output layers.
- NLDL-FR-edRVFL: the FR-edRVFL with the ordinary direct link propagating the non-linear features to the output layers.
- DDL-FR-edRVFL (SedRVFL).

The effectiveness of the proposed DDL was demonstrated by the highest average accuracy amongst the compared models. This could be attributed to two aspects: (1) the propagation of non-linear features, which could be shown by the lower accuracy obtained by LDL-FR-edRVFL where linear features were propagated. This was essential as learning from raw data may suffer from overfitting and the negative impact of noises due to its low SNR; (2) the conditional propagation in DDL, which was shown by the lower average accuracy obtained by NLDL-FR-edRVFL where the ordinary direct link was used. As mentioned in Section 3, the non-linear features propagated may have an

extensive overlapping with the refined hidden features, resulting in a decrease in average accuracy. Hence, dynamic propagation based on the correlation matrix and the discriminant was necessary. More importantly, in Fig. 8, it should be highlighted that the performance of the model showed a step-by-step improvement as the FR block and the DDL were added to the Raw-edRVFL network. The FR-edRVFL showed substantially higher accuracy than the Raw-edRVFL. The performance was further boosted by adding the DDL.

Overall, it could be shown that mapping into super-high-dimension space, which was not covered by the other RNNs, assisted in mining valuable features with high possibilities. The most valuable features were extracted after the FR process. The DDL could further help to boost the performance as it dynamically complemented the spectral information. Therefore, we conclude that the proposed SedRVFL network has a strong feature learning capability to deal with high-dimension EEG data decoding.

5. Discussion

CNN models that have been widely used for EEG decoding have shown that decoding from raw EEG data can be superior in terms of classification performance to using hand-engineering features as the inputs. At the same time, frequency-domain analysis has emerged as a powerful tool for EEG processing (Craik, He, & Contreras-Vidal, 2019) since EEG data are often found to be associated with behavioral patterns (Kandel et al., 2000). On the other hand, for time-domain data, the effectiveness can be compromised by the presence of noise, especially in the context of RNNs where the hidden weights are randomly initialized and remain fixed during training. Specifically, the noises contained in each time step of the input could interfere with the generation of useful hidden features, leading to deteriorated performance. In contrast, the noise contained in the frequency-domain data is typically limited to specific frequency points. The data of the remaining frequency points can help to generate beneficial hidden features. Therefore, using frequency-domain data as inputs in this work could be advantageous in reducing the challenges of decoding raw EEG data.

Compared with the SOTA edRVFL-based networks, the proposed SedRVFL network with improved feature learning capability can better decode high-dimensional and low-SNR raw EEG data. The advantages

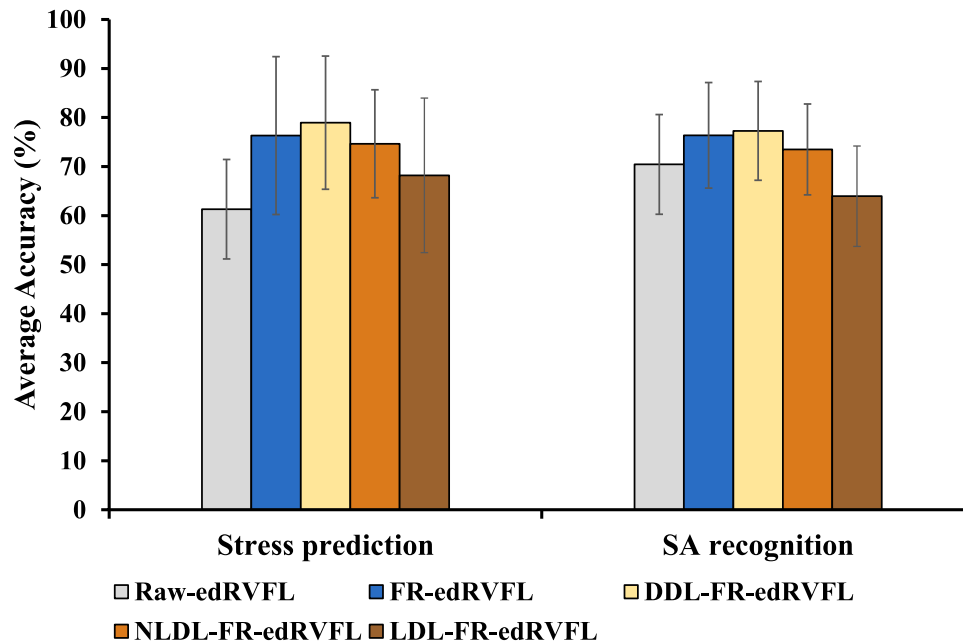


Fig. 8. Average accuracy obtained by applying Raw-edRVFL, FR-edRVFL and its variants.

of the SedRVFL network can be attributed to two aspects. Firstly, the more effective feature refining methods helped guarantee the quality of the extracted features in the hidden layers. By selecting high-variance features (using LVFP and PCA), the FR block was beneficial to not only dimensionality reduction and also to extracting features that were potentially more distinctive. Based on the obtained beneficial features, the classification performance could be boosted. In contrast, the commonly used feature pruning methods (Gao et al., 2023; Shi et al., 2022) are not applicable to process hidden features with over-high dimensionality since the problem of overfitting could limit the generalization capability of the selected features.

Secondly, not only in the hidden layer, the over-high dimensionality of the features propagated from the input space to the output layers was tackled by the DDL algorithm, which was also important for the final classification. Non-linear features were propagated in the DDL. Although the non-linear features may have the disadvantage of only containing limited information, they could help for the dimensionality reduction and complement the features that were undesirable for classification (Shi et al., 2021). Previous studies have also shown that non-linear features such as PSD are useful for EEG decoding (Rojas et al., 2019; Wen & Mohd Aris, 2022). Furthermore, the conditional propagation in the DDL helped to avoid ill-conditioning and instability in the solution when the propagated features and the hidden features were highly correlated. Overall, the DDL algorithm can serve as a regularization for the raw EEG data decoding thereby allowing the SedRVFL network to achieve better performance than the edRVFL variants.

6. Conclusion

This work proposed the SedRVFL network to perform the cross-subject EEG-based pBCI classification tasks. Two enhancement strategies were proposed to deal with the challenging high-dimension data decoding in RNNs. Firstly, the FR-block involved LVFP, PCA and subject-specific standardization to extract features that were more distinctive and to reduce the impact of subject variability. Secondly, the DDL was proposed to employ dynamic propagation of non-linear features to further reduce feature dimensionality and avoid excessive overlapping of feature information. Experiment results showed that the SedRVFL network achieved an outstanding performance over the

strong baselines in stress prediction and SA recognition. The ablation study further demonstrated the effectiveness of the two proposed strategies in the SedRVFL network. In conclusion, the proposed SedRVFL network showed a strong feature learning capability for decoding high-dimension raw EEG data, substantially improving the cross-subject classification performance. This study not only highlights the efficacy of utilizing optimized edRVFL for EEG decoding but also provides a new direction for future research in EEG decoding.

The future works can be considered from three main aspects.

- From the algorithm point of view, although the hyper-parameters of the SedRVFL network were optimized using grid search, there could be a possibility that the optimal hyper-parameters may not appear in the pre-defined search space, leading to sub-optimal model performance. Therefore, better hyper-parameter optimization algorithms such as Bayesian optimization (Pelikan, Goldberg, & Cantú-Paz, 1999) could be explored to further improve the model performance. In addition, more unsupervised feature dimensionality reduction methods that can improve the feature learning capability of edRVFL could be further investigated.
- From the perspective of pBCIs, the application scope of the proposed model can be broadened to various pBCI tasks. Therefore, future works could involve further evaluation of the proposed model on the classification tasks of other mental states such as mental workload.
- From the perspective of practical application, the proposed model could be applied to evaluate the design of next-generation interfaces in dynamic systems. The mental states induced when the operators manipulate the interfaces could be assessed using the proposed model, which could provide references on the design of the interfaces. In addition, against the time-consuming data collection for mental states in dynamic systems, the proposed model can be suitable for small-size EEG-based pBCI datasets in real-world applications.

CRediT authorship contribution statement

Ruilin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft.
Ruobin Gao: Conceptualization, Methodology, Software, Validation,

State Pre-Questionnaire

Please indicate how well each word describes how you feel *At The Moment*.

Not at all = 1	A little bit = 2	Somewhat = 3	Very much = 4	Extremely = 5	
1. Dissatisfied	1	2	3	4	5
2. Alert	1	2	3	4	5
3. Depressed	1	2	3	4	5
4. Sad	1	2	3	4	5
5. Active	1	2	3	4	5
6. Impatient	1	2	3	4	5
7. Annoyed	1	2	3	4	5
8. Angry	1	2	3	4	5
9. Irritated	1	2	3	4	5
10. Grouchy	1	2	3	4	5

Please indicate how true each statement is of your thoughts *During The Past Ten Minutes*.

Not at all = 1	A little bit = 2	Somewhat = 3	Very much = 4	Extremely = 5	
11. I am committed to attaining my performance goals	1	2	3	4	5
12. I want to succeed on the task	1	2	3	4	5
13. I am motivated to do the task	1	2	3	4	5
14. I'm trying to figure myself out	1	2	3	4	5
15. I'm reflecting about myself.	1	2	3	4	5
16. I'm daydreaming about myself.	1	2	3	4	5
17. I feel confident about my abilities.	1	2	3	4	5
18. I feel self-conscious.	1	2	3	4	5
19. I am worried about what other people think of me.	1	2	3	4	5
20. I feel concerned about the impression I am making.	1	2	3	4	5
21. I expect to perform proficiently on this task.	1	2	3	4	5
22. Generally, I feel in control of things.	1	2	3	4	5
23. I thought about how others have done on this task.	1	2	3	4	5
24. I thought about how I would feel if I were told how I performed.	1	2	3	4	5

(a)

State Post-Questionnaire

Please indicate how well each word describes how you felt *During The Task*.

Not at all = 1	A little bit = 2	Somewhat = 3	Very much = 4	Extremely = 5	
1. Dissatisfied	1	2	3	4	5
2. Alert	1	2	3	4	5
3. Depressed	1	2	3	4	5
4. Sad	1	2	3	4	5
5. Active	1	2	3	4	5
6. Impatient	1	2	3	4	5
7. Annoyed	1	2	3	4	5
8. Angry	1	2	3	4	5
9. Irritated	1	2	3	4	5
10. Grouchy	1	2	3	4	5

Please indicate how true each statement was of your thoughts *While Performing The Task*.

Not at all = 1	A little bit = 2	Somewhat = 3	Very much = 4	Extremely = 5	
11. I was committed to attaining my performance goals	1	2	3	4	5
12. I wanted to succeed on the task	1	2	3	4	5
13. I was motivated to do the task	1	2	3	4	5
14. I tried to figure myself out.	1	2	3	4	5
15. I reflected about myself.	1	2	3	4	5
16. I daydreamed about myself.	1	2	3	4	5
17. I felt confident about my abilities.	1	2	3	4	5
18. I felt self-conscious.	1	2	3	4	5
19. I was worried about what other people think of me.	1	2	3	4	5
20. I felt concerned about the impression I was making.	1	2	3	4	5
21. I performed proficiently on this task.	1	2	3	4	5
22. Generally, I felt in control of things.	1	2	3	4	5
23. I thought about how others have done on this task.	1	2	3	4	5
24. I thought about how I would feel if I were told how I performed.	1	2	3	4	5

(b)

Fig. A.1. Short stress state questionnaire (SSSQ) (Helton & Näswall, 2015). (a) State Pre-Questionnaire. (b) State Post-Questionnaire.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See Fig. A.1

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